Networking Basics: LAN (Local Area Network)

A **LAN (Local Area Network)** is a network that connects computers and other devices within a **small geographic area**, such as a home, school, office, or building. It allows devices to share data, printers, internet access, and other resources.

Key Features of a LAN

- **Limited area:** Covers a small location (room, building, or campus).
- **High speed:** Typically offers speeds from 100 Mbps to 10 Gbps or more.
- **Privately owned:** Usually managed by an individual or organization.
- **Resource sharing:** Devices can share files, printers, and internet connections.
- **Low cost:** Less expensive to install and maintain than larger networks.

Components of a LAN

- **Computers/Devices:** PCs, laptops, smartphones, printers, etc.
- **Switch:** Connects multiple devices within the network.
- **Router:** Connects the LAN to the internet or other networks.
- **Network Interface Card (NIC):** Allows a device to connect to the network.
- **Cables or Wi-Fi:** Ethernet cables or wireless access points provide connectivity.

Types of LAN

1. **Wired LAN** – Uses Ethernet cables for fast and reliable communication.
2. **Wireless LAN (WLAN)** – Uses Wi-Fi to connect devices without cables.

Advantages of LAN

- Fast data transfer
- Easy file and printer sharing
- Cost-effective
- Centralised management
- Secure communication within the local network

Disadvantages of LAN

- Limited coverage area
- Requires setup and maintenance
- Hardware failure can affect the network
- Security risks if not properly protected

Example

In an office, 20 computers are connected to a switch and router. Employees can:

- Share files with each other.
- Print using a shared printer.
- Access the internet through the same router.

This setup is a **Local Area Network (LAN)**.

Summary

A **LAN (Local Area Network)** connects devices in a small area, enabling fast communication and resource sharing. It is widely used in homes, schools, offices, and businesses because it is reliable, fast, and cost-effective.